A Report

on

AI-Driven Video Summarization Using LangChain and LLMs

carried out as part of the course Minor Project CS3270

Submitted by

Samagra Gupta
229301588
VI-CSE

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

Department of Computer Science & Engineering,
School of Computer Science and Engineering,
Manipal University Jaipur,
February 2025

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Under the Guidance of :

Guide Name : Mr. Lav Upadhyay

Guide Signature(with date) : ……………………………………

Acknowledgement

This project would not have been completed without the help, support, comments, advice, cooperation and coordination of various people. However, it is impossible to thank everyone individually; I am hereby making a humble effort to thank some of them.

I acknowledge and express my deepest sense of gratitude to my internal supervisor **Mr. Lav Updhayay** for his constant support, guidance, and continuous engagement. I highly appreciate his technical comments, suggestions, and criticism during the progress of this project "**AI-Driven Video Summarization Using LangChain and LLMs**".

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Registration No. 229301588 **Student Name – *Samagra Gupta***

MANIPAL UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)

Department of Computer Science and Engineering

School of Computer Science and Engineering

Date: ______________

CERTIFICATE

This is to certify that the project entitled "***(AI-Driven Video Summarization Using LangChain and LLMs)***" is a bonafide work carried out as ***Minor Project Midterm Assessment (Course Code: CS3270)*** in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by ***(Samagra Gupta)*** bearing registration number**(_229301588_)**, during the academic semester *VI of year 2024-2025.*

Place: Manipal University Jaipur, Jaipur

Name of the project guide: Mr. Lav Upadhyay

Signature of the project guide: _______________________

Contents

1. Introduction

1.1 Objective of the project

The goal of this project is to create a system that can humanize YouTube videos with the help of LaMini-Flan-T5 LLM and Text-to-Speech (TTS) technology. The system is designed to produce short, proportional, and precise summaries of videos to improve user experience and help users manage overwhelming information.

Specific Objectives:

1. In this work, we present an AI-based summarization model that is responsible for the analysis of the YouTube video transcripts with **LaMini-Flan-T5** and the generation of abstractive summaries of high quality.
2. By integrating OpenAI's language model with an AI-driven video summarizer using LangChain, enabling users to ask questions about summarized video content. This integration enhances the system’s interactivity by allowing natural language queries, improving accessibility, and providing insightful responses based on extracted video summaries.
3. To make the interface as simple as possible, we will be using Streamlit, which will allow users to paste video links, read summaries, and hear the audio through the Text-to-Speech API.
4. The system should be able to scale up and adapt to the data so that it can work well with different types of videos and that user feedback should be used to improve the customization of the system.

1.2 **Brief description of the project**

This project uses abstractive summarization with transformer-based models including LaMini-Flan-T5 and GPT4 which are integrated with LangChain to better structure prompts and manage memory to produce short yet detailed summaries of YouTube videos. As extractive methods have problems with maintaining the narrative flow while summarizing content, this approach produces human-like summaries by first understanding the core meaning of the content. Using the YouTube API, the system retrieves video transcripts which are then processed into text and summarized by a Hugging Face pipeline of LLMs. Furthermore, a TTS API is used to turn the summaries into audio so that they are more accessible. A Streamlit-based web interface encloses the entire process and enables users to submit video links which they can preview and receive an AI-generated summary of the video they watched which enhances the content consumption experience with valuable and precise insights.

1.3 **Technology Used**

1.3.1 **Hardware Requirements**

- **Processor**: Intel i5/i7 or AMD equivalent
- **RAM**: 8GB (min), 16GB+ (recommended)
- **Storage**: 20GB free space
- **GPU (Optional)**: NVIDIA CUDA-supported GPU for faster processing

1.3.2 **Software Requirements**

- **OS**: Windows 10/11, macOS, or Linux
- **Python**: 3.8+
- **Libraries**: Hugging Face, LangChain, YouTube API, Streamlit
- **IDE**: VS Code, PyCharm, or Jupyter Notebook

2. Design Description

2.1 Flow Chart

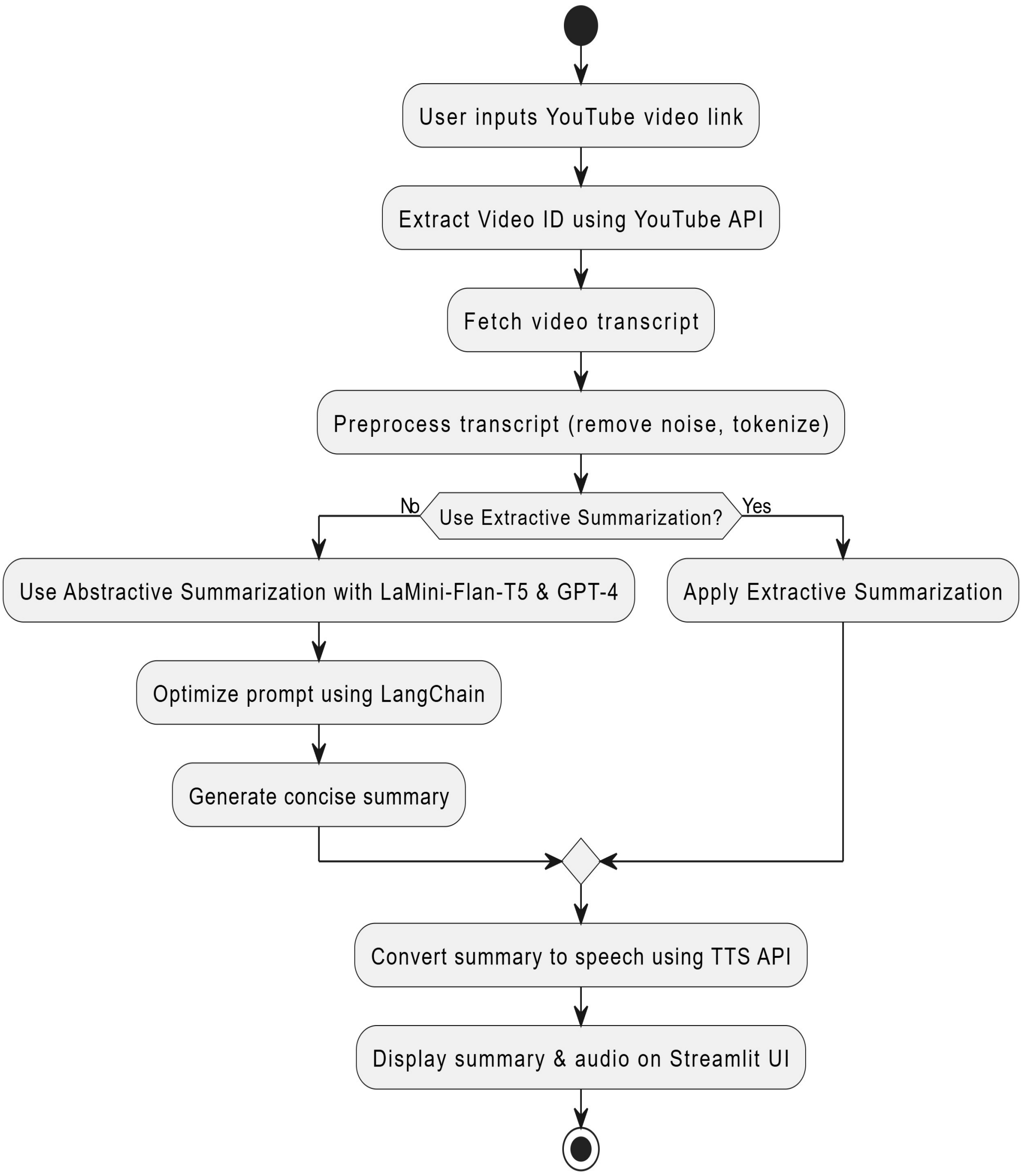

Figure 1. Flow Chart

2.2 **Data flow diagram**

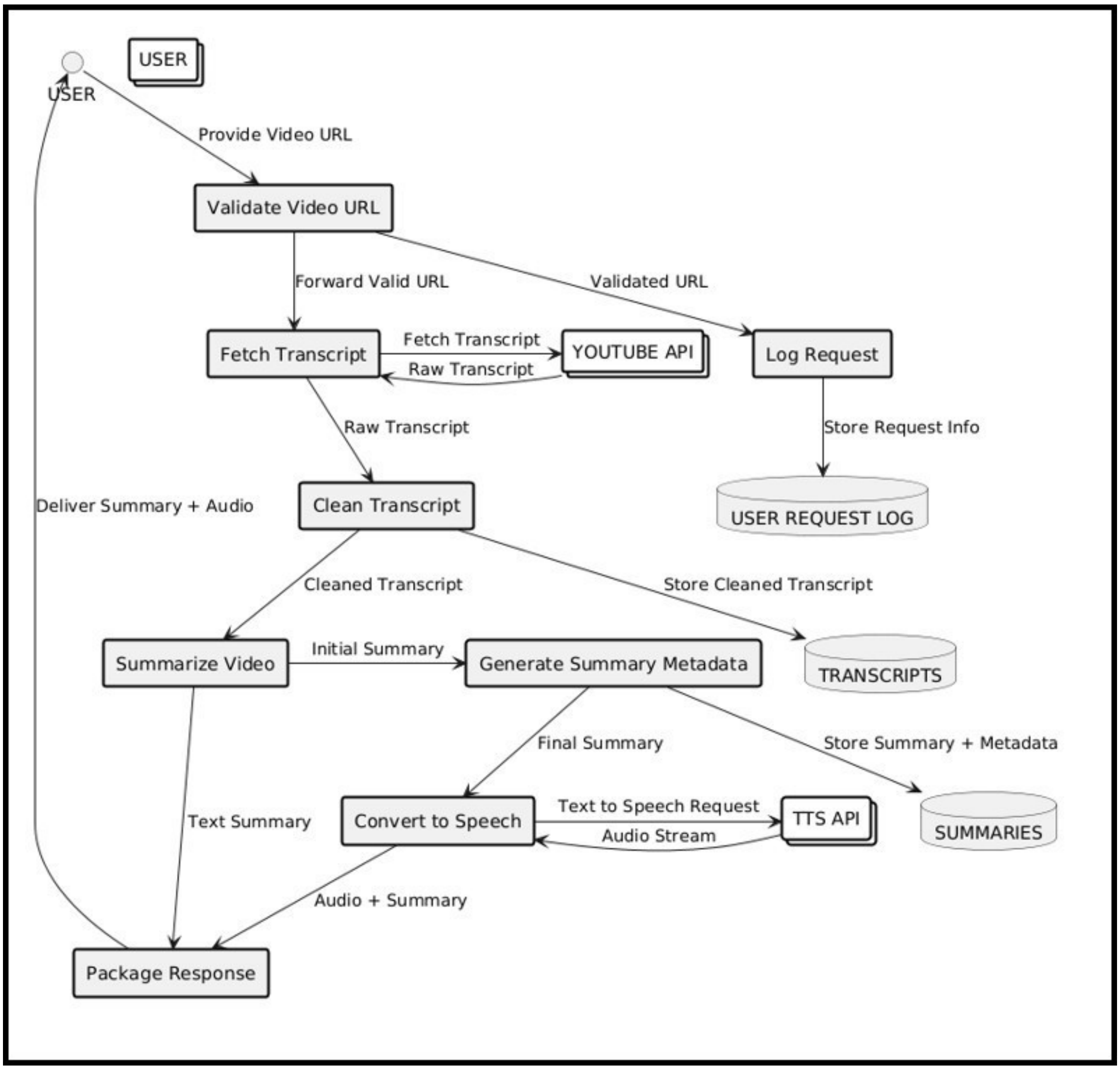

Figure 2. Data Flow Diagram

2.3 **Entity relationship diagram**

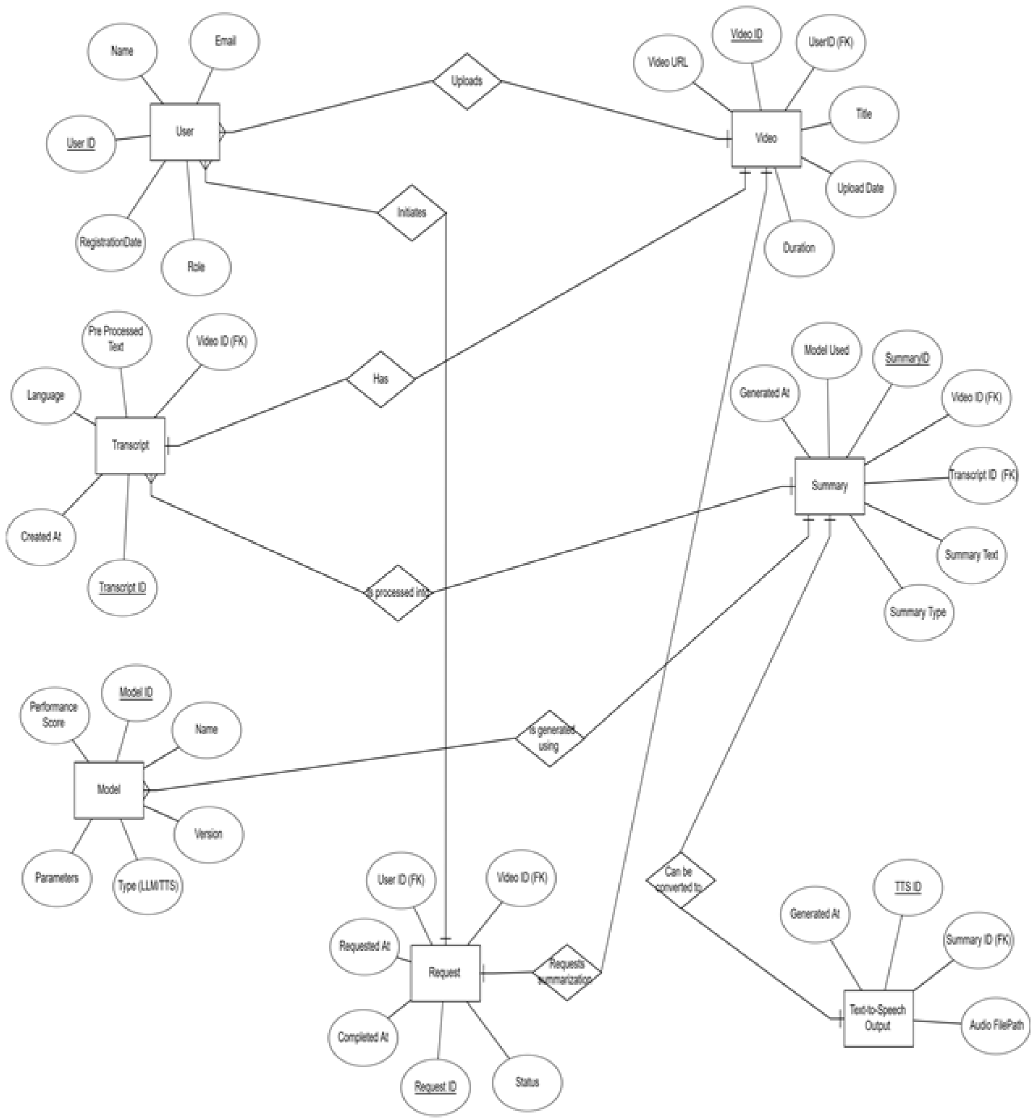

Figure 3. Entity relationship diagram

3. **Project Description**

TVSum Dataset : Title-based Video Summarization (TVSum) dataset serves as a benchmark to validate video summarization techniques. It contains 50 videos of various genres (e.g., news, how-to, documentary, vlog, egocentric) and 1,000 annotations of shot-level importance scores obtained via crowdsourcing (20 per video). The video and annotation data permits an automatic evaluation of various video summarization techniques, without having to conduct (expensive) user study.

4. **Input/Output Form Design**

Input Form Design :

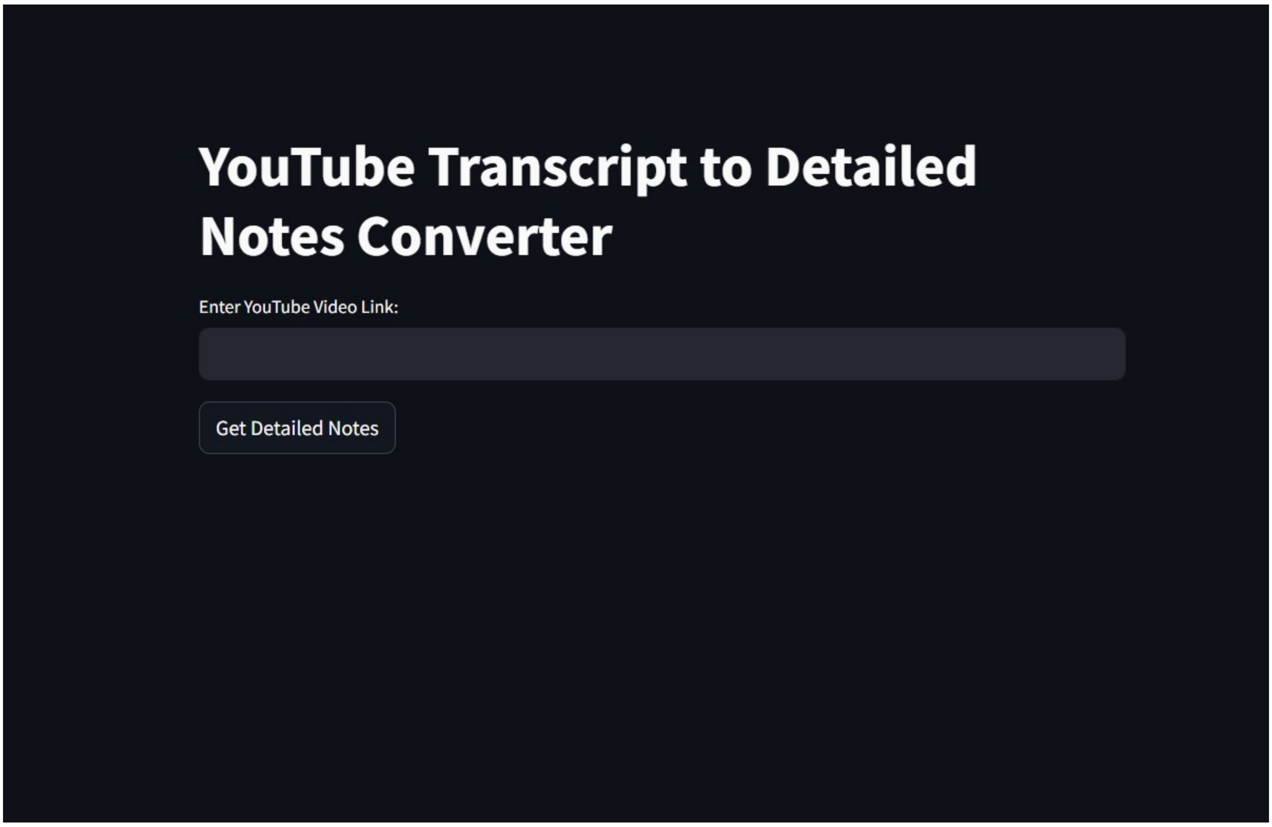

Figure 4. Paste Video URL

Output Form Design :

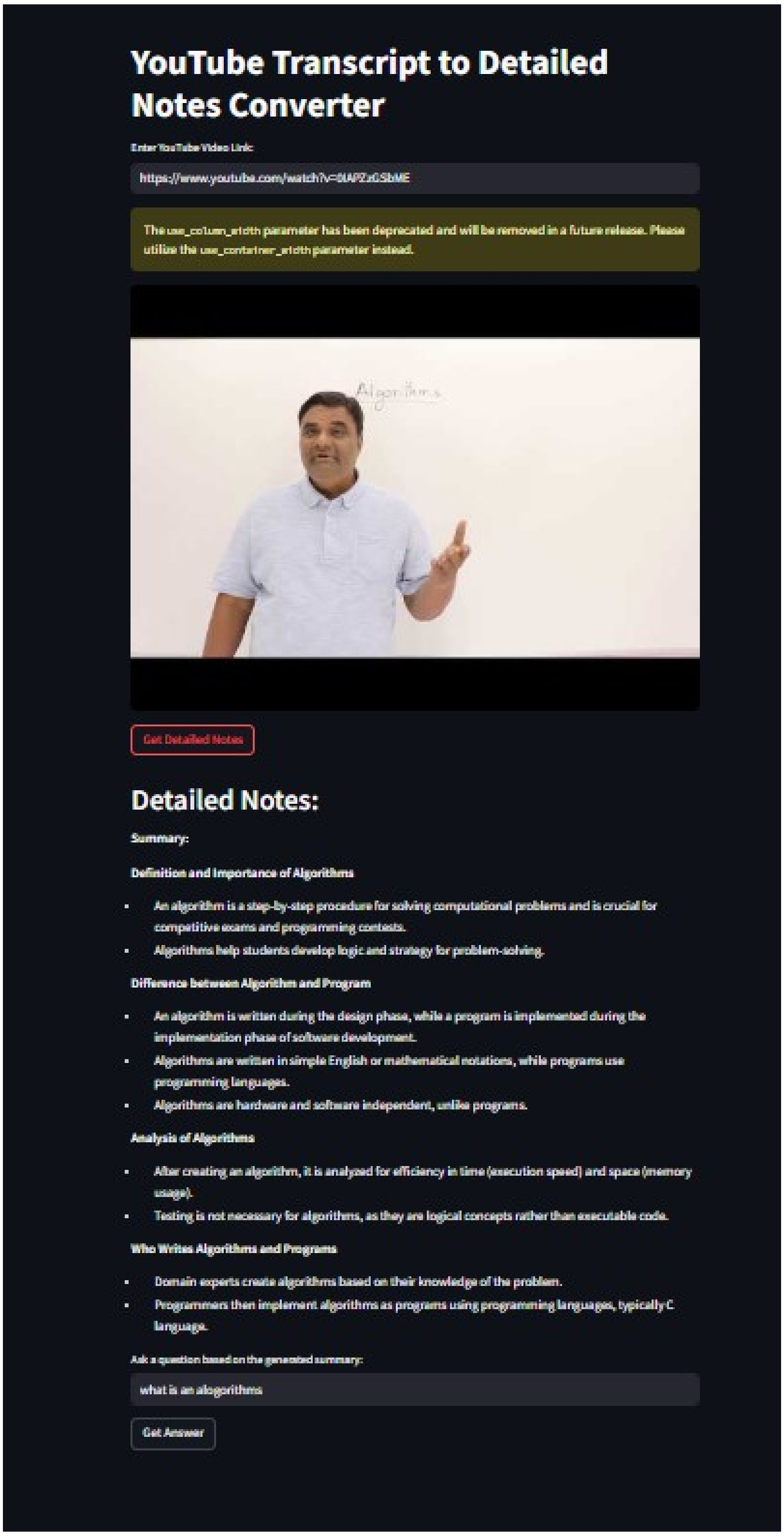

Figure 5. Detailed Summary

Testing & Tools Used

5.1 Testing Approach

A systematic testing plan was used to verify the functionality, accuracy and efficiency of the AI-driven video summarizer. The following testing methods were used:

- Unit Testing: Each module such as video frame extraction, text chunking, vector embedding, and summarization was tested individually to ensure that the expected outputs were obtained.
- Integration Testing: Verified that the LangChain pipelines, ChromaDB and LLaMini-based models interacted smoothly in the summarization workflow
- Functional Testing: Verified that the system could produce accurate and coherent summaries from video transcripts and uploaded documents (e.g. YouTube, PDFs).
- Performance Testing: Tested the execution time and responsiveness, especially for long transcripts and batch video processing.
- Accuracy Testing: The summary output of multiple models (LLaMA-7B, FLAN-T5) was compared using the ground truth from the TVSum dataset.
- Model Comparison Testing: Different models were benchmarked across the same test cases to measure accuracy and reliability.

Tool / Library	Purpose
LangChain	Manages prompt chaining, vector retrieval, and model interaction
LaMini-LLM	Lightweight summarization LLM used for generating text summaries
ChromaDB	Vector store used for embedding and similarity-based retrieval
HuggingFace Transformers	Loads pre-trained LLMs like Mistral, LLaMA, and FLAN-T5
OpenCV	Used for extracting video frames for analysis
Matplotlib & Seaborn	For plotting model accuracy and comparison graphs
Google Colab	Platform used for development, testing, and visualizing results
Whisper (OpenAI)	High-accuracy transcription from audio
TVSum Dataset	Used as benchmark for evaluating summarization relevance and coverage

5. Conclusion and Future Work

5.1 Conclusion

This paper proposes a technique for summarizing videos using AI capabilities which produces abstractive summaries using LangChain and LLMs. We have found that combining prompt engineering and memory management also helps to enhance the fluency, coherence, as well as the contextual relevance of the outputs. However, there are still problems to be addressed, for example, how to address the diverse video content and how to improve the computational efficiency, but there is still much room for improvement from the multi-modal learning technique and enhanced deployment strategies. As for the future work, we will go on to enhance the summarization techniques, real time processing and include the user feedback to enhance the precision and adaptability of the model.

5.2 Future Work

As a starting point for future research, our technique will be elaborated in three simple steps:.

- **Multi-Modal Learning:** Combination of the video and auditory modalities along with the textual information will allow for a multidimensional contextualization and thus lead to dense video minute.
- **Domain-Specific Fine-Tuning:** Our goal is to create tailored AI models which specialize in different video types resulting in more precise and relevant summaries.
- **Real-Time Summarization:** With the help of edge computing and a distributed variant of AI models, we will achieve the real-time, effective live-stream summarization with low latency acceptable and controllable.

By those means we strive to overcome limitations of our summarization system in being more flexible, scalable and generally efficient, for a wide range of applications.

6. Outcome

- Developed an AI-driven video summarization model using LaMini-Flan-T5 for high-quality abstractive summaries from YouTube video transcripts.
- Integrated OpenAI's language model with LangChain, allowing users to ask questions about summarized video content for enhanced interactivity.
- Built a Streamlit-based interface for a simple user experience, enabling users to paste video links, read summaries, and listen to them using a Text-to-Speech API.

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